is SC a good proxy of regions to consider for a PANSS model?

We compare 2 approaches:

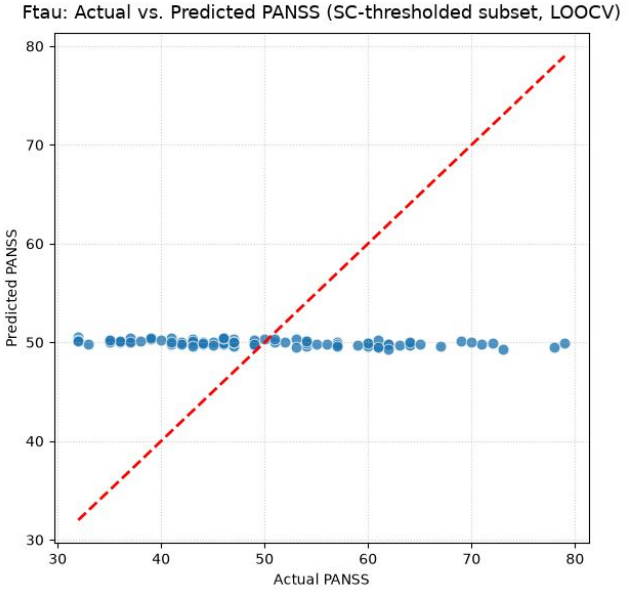
- **SC-thresholded model** (only consider regions that are actually connected, above some threshold) and use Ftau values as features of the model.
- **Ftau-thresholded model** (only consider regions that have a high correlation in the Ftau matrix) and use Ftau values as features of the model. Note: We also tried with the GEC matrix and with different thresholds. A thres=0.2 reported a correlation of 0.56 in actual panss and predicted panss. We also tried different models. A simple ridge regressor was sufficient.

is SC a good proxy of regions to consider for a PANSS model?

Not really. There’s not really an overlap across set of regions. Explicitly: 3 matrix elements in common (Jaccard=0.0060, SC-set size=266, corr-set size=233) under a threshold of 0.2 in the ftau model and a threshold of 0.075 (similar sample size across sets of regions) in the sc-ftau model. Also, SC-Ftau model performs really bad.

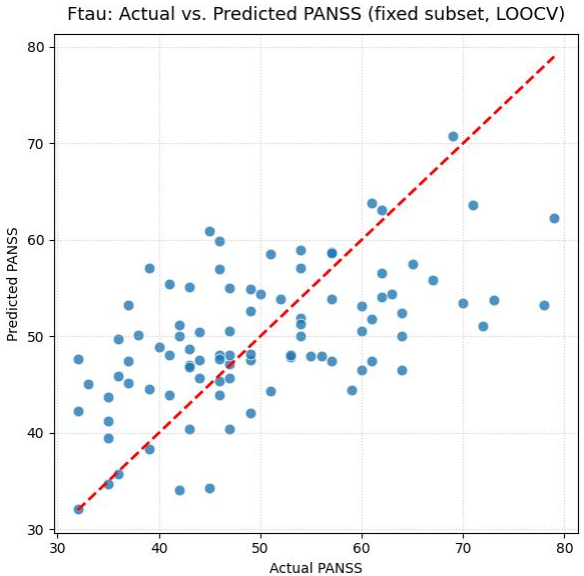
is SC a good proxy of regions to consider for a PANSS model?

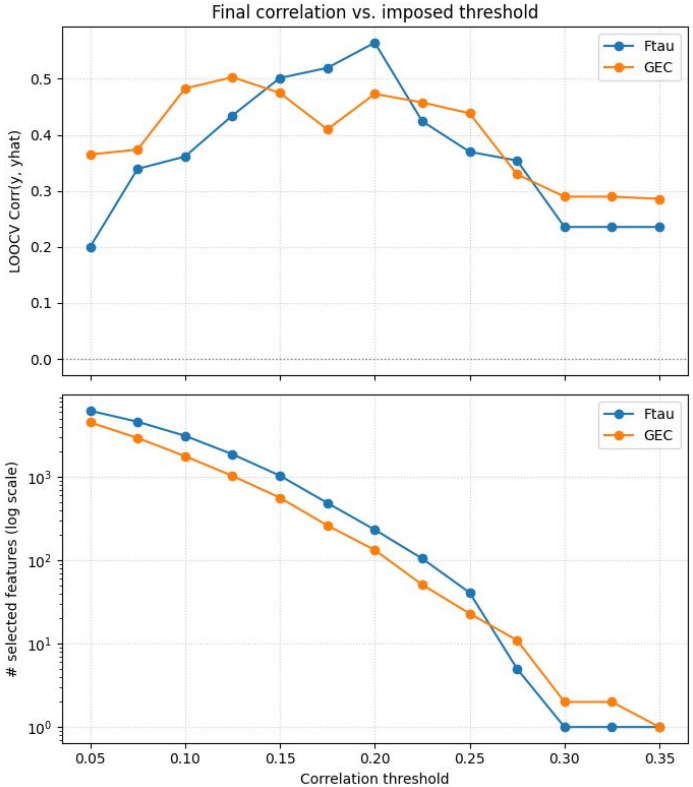
New SC-based model performance. It doesn't learn anything (in basically no correlation threshold)



is SC a good proxy of regions to consider for a PANSS model?

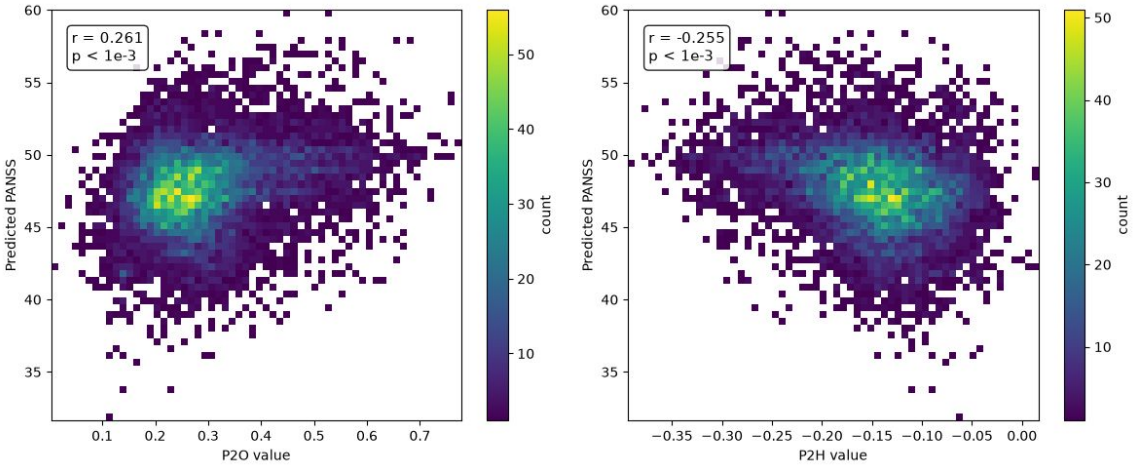
Old Ftau-based model.





Does the perturbation that minimizes P2H or maximizes P2O also minimizes predicted PANSS?

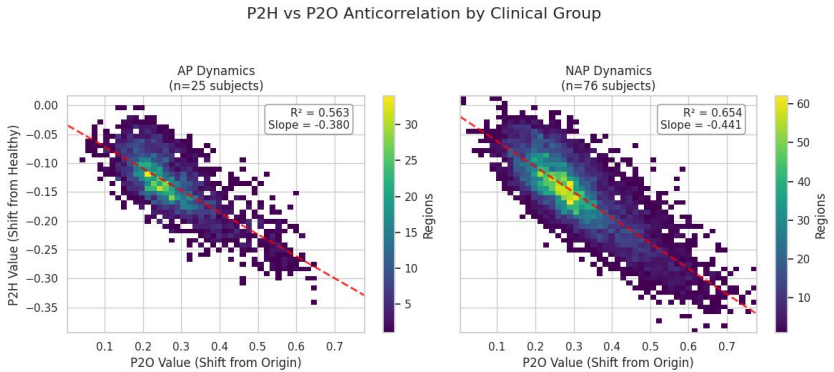
Fixing an Ftau-based model with features correspondent of Ftau matrix elements that correlates > 0.2, we compute a panss score after perturbing each region, individually, for multiple subjects.



Does the perturbation that minimizes P2H or maximizes P2O also minimizes predicted PANSS?

This, to me, reads:

- The greater the pert2origin, the greater the panss.
- The closer pert2healthy to 0, the lower the panss. Were: P2H = sim(FC_pert, FC_healthy) - sim(FC_unpert, FC_healthy).
- We have already seen that P2O and P2H anticorrelate.



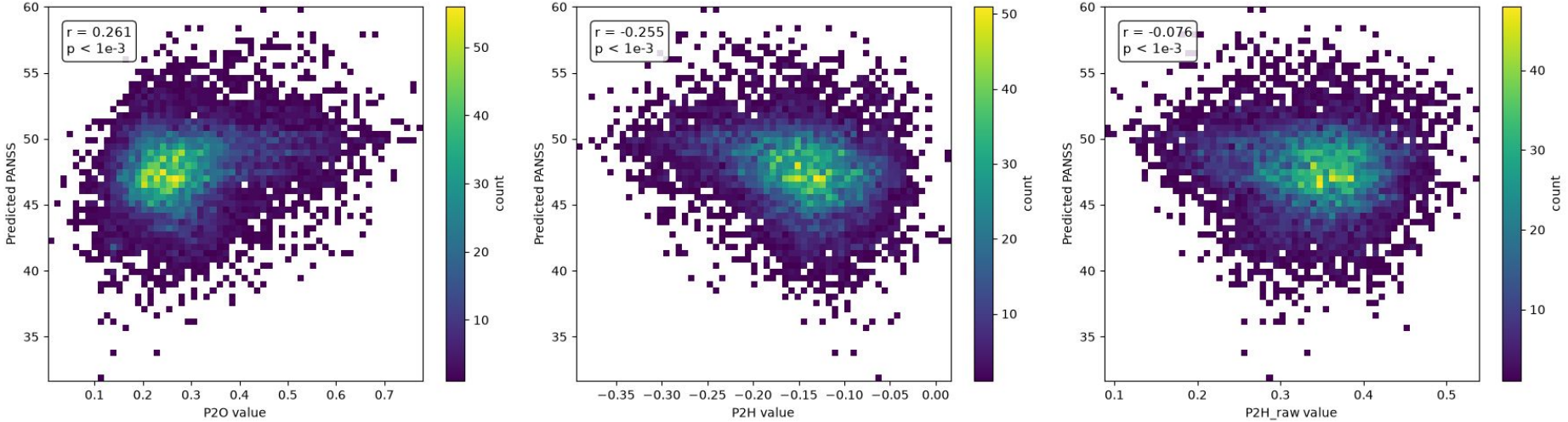
Does the perturbation that minimizes P2H or maximizes P2O also minimizes predicted PANSS?

Questions:

- P2H = sim(FC_pert, FC_healthy) - sim(FC_unpert, FC_healthy). Shouldn't we consider only the first term?
  - If a person is healthy sim(FC_unpert, FC_healthy)=1 and assuming perturbation doesn't helps sim(FC_pert, FC_healthy)=0 → P2H = -1
  - If a person is not healthy sim(FC_unpert, FC_healthy)=0 and assuming perturbation helps sim(FC_pert, FC_healthy)=1 → P2H = 1
  - If a person is not healthy sim(FC_unpert, FC_healthy)=0 and assuming perturbation doesn’t help sim(FC_pert, FC_healthy)=0 → P2H = 0
- It seems that what we want is P2H_new = sim(FC_pert, FC_healthy)

Does the perturbation that minimizes P2H or maximizes P2O also minimizes predicted PANSS?

Assuming a new **p2h_raw** (which is just the first term of the old p2h), we have:



p2h_raw has more variance with r and pearson's rho ~0